1 (a)	$(2, -4)$	B1, B1 **(2)**
(b)	$(8, 6)$	B1, B1 **(2)**
(c)	$(-2, 6)$	B1 **(1)**
		(5 marks)

2. (a)	$f(1) = 4\times1^3 + 2\times1^2 - 12 = -6$ AND $f(2) = 4\times2^3 + 2\times2^2 - 12 = 28$	M1
	States change of sign, continuous and hence root	A1
		(2)
(b)	$4x^3 + 2x^2 - 12 = 0 \Rightarrow x^2(4x+2) = 12$	M1
	$\Rightarrow x^2 = \dfrac{12}{4x+2} = \dfrac{6}{2x+1} \Rightarrow x = \sqrt{\dfrac{6}{2x+1}}$	A1
		(2)
(c) (i)	$\left(x_2 = \sqrt{\dfrac{'6'}{2+1}} = 1.4...\right), \qquad x_3 = 1.2519$	M1, A1
(ii)	$\alpha = 1.2934$	B1
		(3)
		(7 marks)

3. (a)	$\frac{dx}{dy} = \frac{8y(2y+1) - 2\left(4y^2 - 3\right)}{(2y+1)^2}$	M1 A1
	$\frac{dy}{dx} = \frac{(2y+1)^2}{8y^2 + 8y + 6}$	dM1, A1
		(4)
(b)	Sets $\frac{(2y+1)^2}{8y^2+8y+6} = \frac{1}{3} \Rightarrow 4y^2 + 4y - 3 = 0$	M1, A1
	$\Rightarrow (2y-1)(2y+3) = 0 \Rightarrow y = \frac{1}{2}, x = ...$	dM1
	$P = \left(-1, \frac{1}{2}\right)$ only	A1
		(4)
		(8 marks)

4.(a)

$$x^2+0x+1\overline{)3x^3-5x^2+7x-5}\qquad 3x-5$$

$$\underline{3x^3+0x^2+3x}$$

$$-5x^2+4x$$

$$\underline{-5x^2+0x-5}$$

$$4x$$

M1

$\underline{A=3}, \quad B=-5, C=4$ — B1, A1

$D=0$ * — A1*

(4)

(b)

$$\int 3x-5+\frac{4x}{x^2+1}\,dx=\frac{3x^2}{2}-5x+2\ln(x^2+1)+c$$

M1, A1ft

$$\text{Area}=\left(\frac{3\times 3^2}{2}-5\times 3+2\ln(3^2+1)\right)-\left(\frac{3\times 2^2}{2}-5\times 2+2\ln(2^2+1)\right)$$

$$=\frac{5}{2}+\ln 4$$

dM1, A1

(4)

(8 marks)

5.(a)	$\dfrac{dy}{dx} = -10\sin 2x - 24\cos 2x$	M1, A1
	$\left.\dfrac{dy}{dx}\right\|_{x=\frac{\pi}{3}} = -10\sin\left(\dfrac{2\pi}{3}\right) - 24\cos\left(\dfrac{2\pi}{3}\right) = -5\sqrt{3} + 12$	A1
		(3)
(b)	$R = 13$	B1
	$\tan\alpha = \dfrac{12}{5} \Rightarrow \alpha =$ awrt 1.176	M1A1
		(3)
(c)	Sets $-26\sin(2x + 1.176) = 6$	M1
	$\sin(2x + "1.176") = -\dfrac{3}{13}$	A1 ft
	$x = \dfrac{\arcsin\left(-\dfrac{3}{13}\right) - 1.176}{2} =$ awrt 2.44	dM1, A1
		(4)
		(10 marks)

6. (a)	$f'(x) = 6(2x-3)^2 e^{4x-2} + 4(2x-3)^3 e^{4x-2}$	M1 A1
	$= 2(2x-3)^2 e^{4x-2}\{3+2(2x-3)\} = 2(4x-3)(2x-3)^2 e^{4x-2}$	dM1 A1
		(4)
(b) (i)	$x = \frac{3}{2}, \frac{3}{4}$	B1 ft
(ii)	Attempts $f\left(\frac{3}{4}\right) = \left(-\frac{3}{2}\right)^3 e^1 = -\frac{27}{8}e, \Rightarrow g\left(\frac{3}{4}\right) = -27e$	M1, A1
	$-27e \leqslant g(x) \leqslant 0$ g g ç	A1
		(4)
		(8 marks)

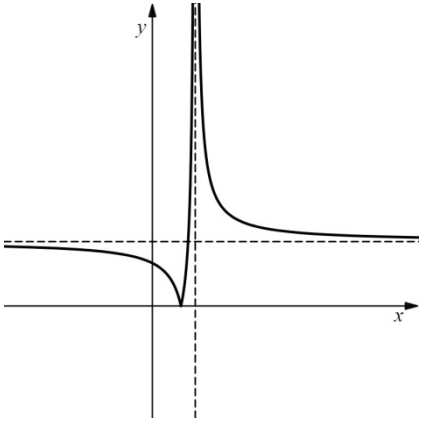
7 (a)	$\sin 4\theta \equiv 2\sin 2\theta \cos 2\theta$	M1
	$\equiv 2\left(2\sin\theta\cos\theta\right)\left(1-2\sin^2\theta\right)$	dM1
	$\equiv 4\sin\theta\cos\theta - 8\sin^3\theta\cos\theta$	
	$\equiv \sin\theta\cos\theta\left(4-8\sin^2\theta\right)$	A1
		(3)
(b)	$\sec x \sin 4x = 5\sin^3 x\cot x$	
	$\frac{1}{\cos x}\times\cos x\sin x\left(4-8\sin^2 x\right) = 5\sin^3 x\cot x$	B1
	$\div\sin x \Rightarrow\ 4-8\sin^2 x = 5\sin^2 x\cot x$	
	$\div\cos^2 x \Rightarrow\ 4\sec^2 x - 8\tan^2 x = 5\tan^2 x\cot x$	M1
	$\Rightarrow 4\sec^2 x - 5\tan x - 8\tan^2 x = 0$ *	A1*
		(3)
(c)	Uses $\sec^2 x = 1+\tan^2 x$	M1
	$\Rightarrow 4\tan^2 x + 5\tan x - 4 = 0$	A1
	$\Rightarrow \tan x = \frac{-5\pm\sqrt{89}}{8} \Rightarrow x = ...$	dM1
	$\Rightarrow x =$ awrt 0.506, 2.08	A1
		(4)
		(10 marks)

8. (a) $g(2a) = \dfrac{3\times 2a - 2a}{2a - a} = 4, \quad f(4) = \ln(2\times 4 - 3) = \ln 5$ — M1, A1 **(2)**

(b) $y = \ln(2x-3) \Rightarrow e^{y} = 2x - 3 \Rightarrow x = \dfrac{e^{y}+3}{2}$ — M1

$f^{-1}(x) = \dfrac{e^{x}+3}{2} \qquad x > 0$ — A1, B1 **(3)**

(c)



Shape of curve to right of $x = a$ — B1

Shape of curve to left of $x = a$ — B1

$y = 3$ marked at correct place on curve — B1 **(3)**

(d) (i) $\dfrac{3\times 8 - 2a}{8 - a} = 10 \Rightarrow a = \ldots$ — M1

$\Rightarrow a = 7$ — A1

(ii) $\dfrac{3x - 2\times 7}{x - 7} = -10 \Rightarrow x = \ldots$ — dM1

$\Rightarrow x = \dfrac{84}{13}$ — A1 **(4)**

(12 marks)

9.(a)	$x = 30\mathrm{e}^{-0.2\times 8} = 6.06\,(\mathrm{mg})$	M1, A1 **(2)**
(b)	$x = 30\mathrm{e}^{-0.2\times 10} + 20\mathrm{e}^{-0.2\times 2} = 17.5\,(\mathrm{mg})$ *	B1* **(1)**
(c)	$30\mathrm{e}^{-0.2\times(T+8)} + 20\mathrm{e}^{-0.2\times T} = 10$ $\left(3\mathrm{e}^{-1.6} + 2\right)\mathrm{e}^{-0.2T} = 1$ $\mathrm{e}^{-0.2T} = \dfrac{1}{3\mathrm{e}^{-1.6} + 2}$ $T = 5\ln\left(3\mathrm{e}^{-1.6} + 2\right) = 4.79$	M1 A1 dM1, A1 **(4)**
		(7 marks)